

Logical Structure as Memetic Fitness: A Non-Cognitivist Account of Moral Consistency

Abstract

We propose a non-cognitivist account of why moral systems converge on logical structure. The mechanism is memetic selection at the scale where personal acquaintance fails: above the threshold Dunbar identifies, cooperation between strangers requires moral codes that can be reconstructed from partial information, applied without central adjudication, and used to make public commitments credible. Logical structure, namely consistency, universalisability, and overridingness, is what makes a moral code do these things efficiently. We integrate Sperber's epidemiology of cultural representations with Dunbar's anthropology of religion to identify the specific selection environment in which logical structure becomes fitness-decisive, then trace its implications. The descriptive/normative dichotomy follows as a consequence: logical structure is not a normative requirement on individual moral preferences but a selected feature of moral systems that succeed in propagation. A corollary is that moral philosophers demanding logical coherence (a demand belonging to ethics, the advocacy of moral codes, not to metaethics like the present account) are participating in the propagation mechanism rather than uncovering normative structure. We offer a new reading of the Frege-Geach problem along the way: rather than a defect non-cognitivists must repair, it identifies a structural feature, the embedding of moral attitudes in deductive form, that moral discourse both exploits for transmission and uses cooperatively to make moral preferences publicly inspectable.

1 Introduction

Moral philosophers routinely demand that moral systems be logically consistent. A person who holds that lying is wrong but also holds, without qualification, that one ought to lie to protect an innocent person is typically told that they face a contradiction, a problem that calls for resolution. The demand for consistency is treated as normatively binding: a moral system that cannot resolve its internal tensions is defective. We argue against this treatment from the standpoint of a non-cognitivist and constructivist account of morality, which is by no means exempt from the consistency expectation. The non-cognitivist standpoint allows us to draw a parallel between our account of preferences (within moral frameworks) and the treatment of preferences in rational choice theory underlying

economics. Our standpoint-relative, constructivist commitment allows us to focus on reasoning about the logical structure of moral statements without making claims about their authority or the validity of their content. In place of a normative justification, we propose a fitness-based justification for the consistency of moral systems.

Moral systems with logical structure, i.e. those that are consistent, universalisable, and whose principles override competing preferences in a stable way, outcompete less structured rivals in the cultural environment. They are easier to transmit, more capable of coordinating behaviour across large and anonymous populations, and more resistant to degradation as they pass between generations and communities. The demand for logical coherence in morality is best understood as a memetic fitness advantage, not a normative constraint. In particular, abandoning the expectation of logical consistency would make moral discourse impossible (or at least meaningless), since logical reasoning is the only tool we have for extending moral commitments and justifying them to others. Abandoning consistency will not, however, absolve people of making choices and having opinions traditionally considered to be under the purview of moral discourse.

The account we develop is an interdisciplinary synthesis. It brings together non-cognitivist metaethics, the epidemiology of cultural representations, the evolutionary anthropology of religion and group size, and empirical moral psychology, disciplines that rarely engage one another, in order to explain a phenomenon each touches only in part.

This paper makes three contributions:

- We show that the logical structure of moral systems serves three functional roles, each a selection pressure rather than a normative requirement: (1) making moral reasoning and discourse possible, (2) pre-commitment, fixing choices in advance so that behaviour stays predictable, and
(3) social propagation, transmitting moral systems across persons and generations and coordinating strangers at scale (sections 3.3–3.4);
- We situate the account among the disciplines it draws on, to which it is largely complementary

rather than opposed, filling the gap between findings each developed for a different purpose: we specify the transmission mechanism by integrating Sperber's epidemiology of representations with Dunbar's findings on group scaling (sections 3.5–3.6), and we explain value capture and the recurring pull toward utilitarianism as the cost side of the same selection pressure (section 3.8);

- We offer a new reading of the Frege-Geach problem: not a deficiency non-cognitivism must repair, but a structural feature moral discourse exploits for transmission and uses cooperatively to make moral preferences publicly inspectable, so deductive embedding is a fitness-enhancing pattern rather than evidence of semantic compositionality (section 4.1).

Here, it is important to make two clarifications. First, the three properties just named, consistency, universalisability, and overridingness, are together what we mean by the logical structure of a moral code. They let it behave like a deductive system, a small stock of general principles from which verdicts on particular cases can be derived, which is what makes reasoning about the code possible in the first place. The kind of logical structure that confers a fitness advantage is not full formal consistency in the logician's sense. Real moral systems are never fully consistent and do not need to be. What matters is that there is enough apparent consistency to support inference to new cases and to let the code compress into a few principles rather than an open-ended list of situations, enough universalisability to apply across the members of a group without requiring case-by-case negotiation, and enough overridingness to resist being traded off against competing preferences in a way that would make the system unpredictable. When we later speak of compressibility (sections 3.3 and 3.8), we mean this consequence of consistency, since a consistent code collapses into a few principles where an inconsistent one cannot.

Second, the argument operates at two distinct levels and applies different claims to each. The first is individual morality, i.e. the set of moral preferences a particular person holds. This set can, in principle, be entirely arbitrary, locally inconsistent across cases, and revisable. We make no claim that it must have logical structure. The only intrinsic constraint is that an individual cannot reason about their own moral commitments deliberately without treating them as if they had logical

structure (as described by point (1) above), and the cost of refusing this is paid by the individual alone.

The second level is society-level morality, which is the partly shared framework circulating in a population. It cannot be specified accurately, since it is a heterogeneous distributed object: different individuals understand the same norms differently, subscribe to them to different degrees, and apply them inconsistently. Whatever moral framework propagates as a partly-shared social object is, however, more likely to succeed in doing so if it has logical structure, even though a sufficiently powerful transmission apparatus can carry a logically unwieldy one.

The two levels are not insulated from each other. Society-level morality exerts continuous pressure on individual morality through the institutions in which individuals participate: families, schools, workplaces, religious communities, legal systems, and media. Individual moral preferences are therefore not only intrinsically free of logical-structure constraints but also shaped, in practice, by participation in institutions that propagate logically-structured moral frameworks. The arbitrary individual morality available in principle is rarely the individual morality observed in practice.

The fitness claim of this paper concerns society-level morality directly, and individual morality indirectly via this institutional channel (as well as via individual reasoning ability). The descriptive/normative dichotomy concerns the societal level. What individuals experience as a normative requirement of logic on their moral preferences is, on our account, the downward pressure of this society-level structure, transmitted through their institutional and social participation, rather than a requirement holding of the preferences themselves. We draw out what this implies for moral philosophy itself, and for the relation between law and morality, in section 3.7.

2 Related Work: Non-cognitivism and the Demand for Logical Structure

The metaethical tradition of non-cognitivism holds that moral statements do not express propositions with truth values, and therefore cannot be straightforwardly true or false. On Ayer's (1936) emotivist account, moral statements express emotional attitudes, e.g. "killing is wrong" is closer to "boo to killing" than to a factual claim about killing. Stevenson (1944) refined this into a richer ac-

count of how moral language functions to influence attitudes in others, not merely to express one's own. Both share the core commitment that moral statements are not in the business of describing moral facts.

Hare's (1952) prescriptivism accepted this broadly but insisted that moral statements carry a logical constraint: universalisability. If I judge that I ought to do X in situation S, I am logically committed to judging that anyone in a relevantly similar situation ought to do X. For Hare, this is not a contingent feature of moral discourse but a requirement built into the logic of "ought." We accept the non-cognitivist starting point but reject Hare's move. Universalisability, we will argue, is better understood as a fitness property of moral systems: one that helps them propagate and coordinate, rather than a logical requirement on moral statements.

The most coherent attempt to preserve logical structure for non-cognitivist moral discourse is Blackburn's (1984) quasi-realism. Blackburn starts from an expressivist base: moral statements express attitudes. However, he argues that through a process of earning the right to talk as if moral statements were truth-apt, we can recover the logical behaviour of moral terms, including their embeddability in inferences. The Frege-Geach problem (1960, 1965) concerns the difficulty a non-cognitivist faces in accounting for how a moral statement keeps a constant meaning when it passes from assertion into an unasserted context like the antecedent of a conditional. The validity of moral inferences depends on this, and, on Blackburn's account this is a problem to be solved rather than dissolved.

We part ways with Blackburn here deliberately. The Frege-Geach problem is real, and we do not try to solve it. Moral statements are preferences, and preferences do not embed in logical inferences the way propositions do. The appearance that morality must be logically structured, the intuition that drives both Hare's universalisability constraint and Blackburn's quasi-realist project, is what we set out to explain, and our explanation will be memetic rather than normative. We return to the Frege-Geach problem in section 4.1.

Schroeder (2008) offers another approach to recovering logical inference for non-cognitivists. His bifurcated attitude semantics assigns "being for" attitudes to act-types and derives inference pat-

terns from the logical structure of these attitudes rather than from propositional content. Like Blackburn's, Schroeder's project is reconstructive, aiming to vindicate the apparent logical behaviour of moral discourse from an expressivist base, and it meets the same response we gave Blackburn: what both projects take as given, that moral discourse must support logical inference, is exactly what we take as the explanandum.

The closest ancestor of the present account is Gibbard's (1990) norm-expressivism. Gibbard combines a non-cognitivist base, on which moral and normative judgements express states of accepting systems of norms rather than beliefs about normative facts, with an explicitly evolutionary rationale: the human capacity for normative governance, and for the normative discussion that coordinates action and emotion, is on his account an adaptation for social coordination. This pairing of non-cognitivism with a coordination-first evolutionary story is the nearest precedent for what we do, and our account inherits its spirit. We diverge on two points. First, Gibbard's evolutionary explanation targets the purpose: why our species came to make normative judgements and argue over them at all. Ours targets the cultural selection of moral systems: why the frameworks that propagate across populations converge on consistency, universalisability, and overridingness. These are different explanations at different levels, and the second does not follow from the first. Second, Gibbard develops a logic of norm-acceptance precisely in order to vindicate the consistency and embedding behaviour of normative judgement, treating that behaviour as constitutive of the attitudes involved. In this respect he belongs with the reconstructive project just set aside: the logical structure his semantics is built to deliver is, on our account, the very thing to be explained rather than assumed.

Constructivists like Rawls (1971) and Korsgaard (1996) argue that moral norms are constructed rather than discovered in some mind-independent moral reality, which aligns with our commitments. However, Rawlsian construction yields principles that are genuinely normatively binding on rational agents that have chosen to hold them, and Korsgaard's constitutivism grounds moral obligations in the conditions of rational agency itself. For both, construction produces something with cognitive content and normative force. Our constructivism is non-cognitivist: moral systems are built rather than discovered, but what gets built are preferences, not facts.

Nevertheless, our points of view share a desire for an agent to have their moral code propagate: to be adopted, acted upon, and transmitted across persons and generations. Where we differ is in thinking that this propagation ambition is prior to and independent of the logical structure they impose. The logical structure is recruited in service of propagation. This applies even to constructivist moral codes that make non-imposition a central virtue. A constructivist who holds that one ought not impose one's moral preferences on others still, ideally, wants others to accept this meta-moral principle and adopt the view that non-imposition is a virtue. In advocating for tolerance, the constructivist is propagating a moral code that includes tolerance as a value. There is no exit from the memetic dynamic: even a moral framework designed to resist imposition is itself subject to propagation pressures, and the constructivist who holds it participates in those pressures whether or not they acknowledge doing so.

A closer ally is Prinz (2007), whose sentimentalist account holds that moral concepts are emotionally grounded responses calibrated by cultural learning rather than responses to mind-independent moral facts. Moral judgements, on Prinz's account, are culturally variable in content because the emotional responses they express are shaped by the social environment. This view, again, aligns with our non-cognitivist starting point. Where we go further is in explaining why moral systems, despite variable content, tend to converge on the same structural properties. Prinz's sentimentalism leaves this convergence unexplained.

The closest contemporary methodological ally is Thomasson (2026), whose function-first metaethics explains the features of moral discourse by the functions that discourse performs rather than by correspondence to moral facts, and who reconstructs those functions through empirical linguistics, preserving the grains of truth in expressivist, prescriptivist, and descriptivist views. We share the function-first orientation but arrive at it from a different direction and answer a different question. Thomasson traces how the several functions of moral language are layered synchronically across its grammatical forms. We ask instead why one of those functions, transmission and coordination at scale, exerts over time a selection pressure that drives moral systems to converge on logical structure. The two projects are complementary: a linguistic analysis of what

moral discourse now does, and a cultural-evolutionary account of why the systems that survive propagation acquired that structure.

A cluster of naturalist accounts of morality lies close to ours and should be distinguished from it. Skyrms (1996) and Kitcher (2011) approach moral norms as evolved devices of social coordination. Skyrms shows that certain normative structures stabilise in populations through iterated game-theoretic dynamics, not because they are correct but because they are equilibria, so that no individual agent has reason to deviate unilaterally. Kitcher frames morality as a long-running collective human project of improving social coordination, with each generation inheriting and revising a moral code shaped by the problems their predecessors faced. Both accounts are compatible with ours, though neither draws the memetic fitness conclusion we draw.

Sterelny (2012, 2021) develops a sustained naturalist account of how moral norms scaffold cooperation, emphasising apprentice learning and collective intentionality as the mechanisms by which moral norms are transmitted, applied, and revised across generations. Where Sterelny grounds moral norms in cooperation-scaffolding via the cognitive and social machinery available to small-scale human groups, we focus on the specific structural properties (consistency, universalisability, overridingness) of moral systems that succeed at the larger scale Dunbar identifies. These accounts are compatible but distinct. The fitness pressure for logical structure is, on our account, decisive precisely at the threshold where Sterelny's apprentice-learning mechanisms run out, which is the same threshold Dunbar identifies (section 3.6).

Curry's morality-as-cooperation theory (Curry 2016; Curry et al. 2019) is a close relative that must be distinguished carefully. Curry argues that morality just is a collection of biologically and culturally evolved solutions to the recurrent problems of cooperation, namely kinship, mutualism, exchange, and the rest, and marshals cross-cultural evidence that these cooperative goods are everywhere treated as morally good. This is a thesis about the content and origin of moral norms: which norms exist, and why those and not others, while ours is about their structural form. The two claims are orthogonal and mutually supporting: if moral norms are cooperative solutions, then the coordination environment in which they must be transmitted is exactly the one in which, on our

account, logical structure becomes fitness-decisive (section 3.6).

Girard's mimetic theory (Girard 1977) offers a parallel account worth noting. Where Dunbar explains moral codes as coordination technology, Girard sees them as violence-containment technology, i.e. shared prohibitions that channel and redirect the mimetic rivalry that would otherwise tear communities apart. The two accounts are not in competition: a moral code can simultaneously reduce coordination costs and contain mimetic violence, and both functions favour the same structural properties of universalisability, consistency, and overridingness. Girard's account adds a further dimension to why these properties are fitness advantages: they are not only cheap to transmit but actively necessary for social stability.

A final interlocutor worth situating ourselves against is Parfit (1984), whose project in *Reasons and Persons* uses game-theoretic structure to constrain which moral theories are viable. Parfit argues that any moral theory which is self-defeating, i.e. where agents following it produce worse outcomes by the theory's own lights than agents who don't, is thereby undermined as a candidate for the correct moral view. This is a sophisticated use of preference-ordering and collective action logic to do normative work: the structure of rational choice theory is imported into ethics to determine what morality must look like.

We find the analytical machinery illuminating but reject the normative ambition. Parfit is using game-theoretic structure to ask which moral systems are correct. We are using something structurally similar to ask which moral systems propagate and persist. Systems that are self-defeating in Parfit's sense are also likely to be poor competitors in a memetic selection environment. In our view, however, viability under collective action is a fitness property, not a requirement. More broadly, imposing a game-theoretic preference ordering on moral preferences assumes that moral preferences must behave like the preferences of rational agents in a formal decision theory, i.e. transitive, consistent, maximising. We abandon this assumption. Moral preferences are preferences, and like all preferences they can be revised, contextual, and locally inconsistent without anything having gone wrong.

3 The Argument

3.1 Where Logical Structure Actually Binds

Before arguing that morality does not require logical structure, we should be precise about where logical structure does have genuine binding force. Formal systems with explicitly defined terms are governed by a specific logic from which new statements can be derived. In arithmetic, for instance, “ $1 + 1 = 2$ ” is not an empirical discovery but a consequence of how the terms are defined. The binding force is entirely internal to the system.

Descriptions of physical reality are different in kind. We cannot prove from first principles that reality conforms to any particular formal system. Instead, we observe patterns and construct mathematical models to represent them. Those models can be tested and refined, and some achieve extraordinary accuracy, but they derive their authority from empirical fit rather than internal definitional necessity.

Moral statements have neither anchor. They are not formal variables defined into a system whose internal logic then constrains what follows. And there are no moral facts readable off the laws of physics: moral claims are not falsifiable by observation of the world. The demand for logical consistency in morality therefore cannot borrow its authority from either source. It must be justified on other grounds, or not at all. We argue it cannot be justified as a normative requirement, and that the other grounds are entirely functional.

3.2 The Pragmatic Tension

If we fully admitted, at the level of ordinary moral practice, that morality need not be logical, we would lose the ability to do moral reasoning at all. Logical reasoning is the only cognitive tool we have for thinking carefully about moral questions, including deriving judgements about novel cases from existing commitments, identifying conflicts between moral attitudes, and publicly justifying moral positions to others. A moral discourse that explicitly abandoned logical structure would collapse into an enumeration of situation-specific preferences with no generative power.

This is not a refutation of our central claim. It is an illustration of how thoroughly the fitness advantage of logical structure has shaped even our meta-level thinking about morality. We cannot reason about morality without a framework, because the framework is the only reasoning tool available. The purpose of a logical moral framework is, therefore, for use in performing cognitive and social work that its structure makes possible: extending moral attitudes to new cases, resolving apparent conflicts, and, above all, transmitting moral commitments in a form that others can engage with, evaluate, and adopt. The pragmatic usefulness of logical structure in these situations is one of the reasons for its propagative prowess, not evidence of its normative necessity.

A commonplace attitude in philosophy holds that one must follow one's premises wherever they lead. Our account argues against this. The familiar transplant case makes the tension vivid: a surgeon could save five dying patients by killing one healthy visitor and distributing their organs, and the aggregate-welfare calculus that recommends this is notoriously difficult for utilitarians to answer on their own premises. Almost no one accepts the conclusion, and our account explains why they are not obliged to. A moral code that licensed such killings would wreck the very coordination it exists to secure, since no one could trust a hospital. The revulsion that blocks the inference is not a lapse of rigour to be corrected by cleaner logic, but the selection pressure doing its work. Following an argument to its conclusion is warranted only where the conclusion serves the propagative and coordinating function of the code. Where it would corrode that function, declining to follow it is fidelity to what moral reasoning is for, not a failure of it.

3.3 The Three Functional Roles of Logical Structure

We now argue that logical structure in morality serves three distinct functional roles, ordered by their scale of operation and by their dependence on one another. Each role explains why consistency is useful without implying that it is normatively required.

Role (1) is making moral reasoning and discourse possible (see section 3.2). Without enough logical structure to support inference, there is no moral philosophy, no public moral argument, and no way to extend moral attitudes to cases not yet encountered. This operates at the level of individual

cognition and is a precondition for the other two roles.

Role (2) is pre-commitment. Agents use consistent moral frameworks to decide in advance how they would respond to situations they have not yet faced, which lowers the deliberative cost of novel cases and gives others a stable basis for expectation. On a non-cognitivist reading this runs parallel to the treatment of preferences in decision theory and rational choice: because moral preferences are taken as given rather than as truth-apt, they behave like the preferences economics models, where transitivity and completeness are imposed not because preferences are found to be consistent but because a consistent ordering gives economics a solid non-contradictory model of human behaviour on which to build other research. Consistency there is an idealisation that earns its place by doing coordinating and predictive work, not a normative truth about what anyone must prefer, and our claim about moral preferences has the same form. The mistake is to import the requirement into morality normatively, treating moral preferences as those of a rational agent in Savage's framework (Savage 1954). This is because pre-commitment can be violated. For example, someone who pre-commits to preferring A over C, having preferred A over B and B over C, may still prefer C when the choice arrives, because the context shifted or they simply changed their mind, and nothing has gone wrong.

Role (3), social propagation, is the most consequential of the three. A moral system compressible into a small set of principles outcompetes more loosely organised rivals in the cultural selection environment: it is cheaper to transmit, its violations generate visible contradictions that inconsistent codes do not, making defection easier to challenge and harder to ignore, and it makes the expressed attitudes of strangers predictable enough to lower the trust costs of cooperation across anonymous populations (Dunbar 2022, 1996). This role carries the main weight of the account, and its two halves are developed in the two sections that follow: why logically structured codes are cheap to reconstruct and resistant to degradation, the transmission mechanism, in section 3.5, and the threshold beyond which coordinating strangers makes that advantage decisive, the selection environment, in section 3.6.

3.4 Reasoning as Propagation, not Deliberation

The empirical case for our central claim, that moral reasoning serves propagation rather than action-guidance, rests on two converging lines of evidence. The first is Mercier and Sperber's account of reasoning (Mercier and Sperber 2017), developed below, on which reasoning evolved as a social faculty for producing and evaluating arguments to persuade others, not as a tool for private deliberation or truth-seeking. The second is Haidt's social intuitionist model (Haidt 2001, 2012), on which moral judgements are reached by rapid intuitive processes and reasoning enters in their service. Haidt's experimental evidence is the phenomenon of moral dumbfounding, where subjects presented with harmless taboo scenarios, such as consensual sibling incest that harms no one, maintain firm condemnation even after every reason they offer is answered, conceding that they cannot say why they hold the judgement but holding it nonetheless (Haidt et al. 1993, 2000). The taboo example is illustrative of logic being the medium through which preferences of any origin, whether emotional, instinctual, or evolutionary, enter public moral discourse and become available for transmission.

The conclusion to draw from this evidence, that logical moral reasoning is often done to support a previously made decision, is that the function of logical structure is, at the very least, not just action guidance. We argue that function is social and propagative, since reasoning is how moral attitudes get articulated, defended, shared, and passed on, not only how they get applied. This propagative role is what one expects from a faculty that, on Mercier and Sperber's account, evolved for persuasion rather than truth-tracking, and the post-hoc patterns Haidt documents are its signature.

The main qualification comes from Greene's dual-process account (Greene et al. 2001; Greene 2013), on which a slow deliberative system can override fast emotional intuitions in reflective conditions and is genuinely action-guiding in some cases, particularly the novel or high-stakes dilemmas where intuitions are absent or conflicting. Other work points the same way: moral-grammar research attributes computational structure to fast judgement (Mikhail 2011), Cushman's model-based and model-free distinction leaves room for deliberation (Cushman 2013), and Greene's own later work softens his strongest version (Greene 2014). We need not resist any of this.

Our argument needs only the weaker claim, robust across that literature, that reasoning's primary social function is propagation rather than private deliberation, since selection on logical structure operates whether moral reasoning is post-hoc, deliberative, or both. Even in the novel cases Greene emphasises, where no prior intuition fixes the answer and, for a non-cognitivist, no fact of the matter settles it, what confers a judgement's moral standing in a community is adherence to logical reasoning and the propagation of the resulting argument. This is why explicit reasoning is most active precisely where behaviour is least fixed by prior intuition, in unfamiliar dilemmas, public justification, and transmission across generations and groups, the contexts where propagation matters most: in the absence of moral intuition, the logically consistent articulation the immediate persuasion task already demands is both the basis for the moral decision and the very thing that lets a moral code travel.

3.5 The Selection Mechanism: Epidemiology and Cultural Attraction

The concept of the meme, introduced by Dawkins (1976) in *The Selfish Gene*, describes a unit of cultural transmission that propagates through imitation and communication. Like genes, memes are subject to selection pressures: those that are easier to transmit, more memorable, more generative, and more resistant to corruption tend to outcompete those that lack these properties. The more empirically developed statement of that machinery is the dual-inheritance theory of Richerson and Boyd (Richerson and Boyd 2005), on which culture is a second inheritance system with population dynamics of its own. We take the selection-pressure reasoning from that tradition without committing to any particular unit of transmission.

Sperber's (1996) epidemiology of representations offers a more precise framework for thinking about why some cultural items spread more successfully than others. On Sperber's account, cultural representations propagate when they are easy to reconstruct from partial information, fit well with existing cognitive dispositions, and generate further representations when processed. A moral system with logical structure has exactly these properties: its rules can be reconstructed from examples, they fit with the general human tendency to seek consistent frameworks for navigating the social world, and they generate new moral judgements when applied to novel situations. A moral

system without logical structure requires something closer to memorisation of situation-specific rules. This is more expensive to transmit and more vulnerable to degradation across generations. This is the transmission mechanism named by role (3) of section 3.3.

A methodological caveat is due here, because Sperber's own research tradition has moved away from the Dawkinsian framing we have so far borrowed. Cultural attraction theory (Claidière et al. 2014; Scott-Phillips et al. 2018) holds that cultural items are rarely copied with the fidelity a replicator model requires: they are reconstructed anew by each recipient, and recur across a population because transmission is biased toward stable end-points, or attractors, rather than because high-fidelity copies are differentially selected. Our argument does not depend on the replicator reading. What we have claimed is that logically structured moral systems are cheap to reconstruct from partial information and resistant to degradation in transmission. This is, in the theory's own terms, to say that logical structure is an attractor: the form toward which moral systems are repeatedly reconstructed, because it is what fragmentary exposure most readily and stably regenerates. Claidière, Scott-Phillips, and Sperber (2014) show that the attraction and selection frames coincide under specifiable conditions, so the descriptive/normative conclusion we draw does not turn on the choice between them. We retain the fitness idiom for its familiarity, intending it throughout to be neutral between selection among variants and convergence on an attractor.

3.6 The Selection Environment: Scaling and the Stranger Threshold

The most direct empirical grounding for our memetic fitness argument is Robin Dunbar's anthropological work on the evolution of religion (Dunbar 2022, 1996). Dunbar argues that religion evolved primarily as a group-cohesion technology, allowing cooperation in groups larger than the roughly 150 stable relationships individual cognition can sustain (Dunbar's number). Beyond that threshold, coordination requires external scaffolding, which religion supplies through shared ritual and, crucially, its moral codes.

Dunbar distinguishes the religions of small hunter-gatherer communities, which bond members through shared experience and lack formal moral codes, from the doctrinal religions that emerge

as groups grow larger, marked by a moralising high god, formal institutions, and a theologically justified moral code that regulates an otherwise anonymous population. That code is central rather than incidental, since a shared framework makes the behaviour and expressed attitudes of strangers predictable enough to support cooperation. The predictability is only partial (Haidt 2001), since moral reasoning is often post-hoc rationalisation rather than a genuine guide to action, but it need not be complete to do its coordinating work.

What Dunbar adds to the memetic framework of Dawkins and Sperber is the selection environment itself. Below Dunbar's number, personal acquaintance, reputation, and the shared moral intuitions Haidt documents (Haidt 2001, 2012) are enough to regulate behaviour. Above it, one must cooperate with strangers, adjudicate disputes without shared history, and transmit norms to people one will never meet, and only abstract, codified, universalisable rules can do this. The fitness advantage of logical structure is therefore not uniform but becomes decisive precisely where the intimacy of small-group life gives way to the anonymity of large-scale society. This is the selection environment that role (3) of section 3.3 names.

Norenzayan's (2013) account in *Big Gods* provides direct evidence for the specific mechanism at work. Norenzayan argues that belief in moralising, omniscient, punishing deities enabled large-scale cooperation by making moral commitments credible to strangers. Where small-group religion relies on shared ritual and personal acquaintance to sustain social bonds, the moralising high god functions as a coordination technology: an agent believed to monitor behaviour universally and punish defection regardless of social proximity. The universalisability of the associated moral code and its applicability to strangers who cannot be monitored personally are crucial features of its social function. Norenzayan's mechanism converges closely with Dunbar's work on religion and the scaling of human groups discussed above.

This mechanism also explains why the institutional apparatus matters so much. Dunbar's account of how major world religions propagated their moral codes points to the role of priestly hierarchies, centralised bureaucracies, and formal rituals in sustaining transmission across generations (Dunbar 2022). A logically structured moral code can propagate further and more reliably with such an

apparatus behind it. Secular moral systems lack the church, the priesthood, and the ritual. What they have are philosophy, education, legal institutions, workplaces, and media. None of these carry the kind of binding authority that historical religious institutions did, particularly not the credibility of supernatural monitoring that Norenzayan (2013) argues was central to enforcing moral commitments across strangers. In the absence of comparable institutional authority, logical structure is what a secular moral meme has to rely on almost entirely.

The closest formal cousin to one strand of our account is the game-theoretic model of Lie-Panis, Fitouchi, Baumard, and André (2026). They ask why societies bind themselves to rigid moral rules, categorical obligations and prohibitions applied even where case-by-case reasoning would do better, and answer with an efficiency-robustness trade-off: because the reasons people have for failing to cooperate are typically opaque to observers, a flexible norm that admits excuses is open to exploitation, whereas a rigid norm forecloses the loophole at the cost of some efficiency. Their model predicts rigidity precisely where flexibility is riskiest: in low-trust settings or among strangers. This bears on our account in two ways. Its central prediction independently corroborates the threshold argument we take from Dunbar, arriving at the strangers-versus-intimates boundary from formal first principles rather than anthropology, and it supplies a strategic mechanism for one of our three properties: overridingness, the refusal to trade a moral commitment off against competing considerations, is on their analysis what makes a rule robust to disguised defection.

Our model differs in level and in scope. Their model selects equilibria among strategically rational agents inside a cooperation game, while ours concerns the cultural selection, or attraction, of whole moral systems as they propagate across persons and generations, and it assumes no such rationality of the carriers (section 3.4 denies it). Where they explain rigidity, we treat overridingness as one of three logical-structural properties, alongside consistency and universalisability, whose joint convergence the transmission environment explains. Their result is thus a formal deepening of one strand of the present account at the strategic level.

Henrich's work on the cultural evolution of moral norms (Henrich 2020) supplies empirical grounding for much of the propagation story. His findings on how the medieval Western Church's marriage

and family rules, by dissolving extended kin networks over generations, reshaped the psychology of the populations subject to them (the WEIRD profile) bear directly on the institutional-apparatus argument above. They document a historical case in which a moral framework, enforced by a coordinating institution, restructured not merely the beliefs but the moral psychology of a whole population, which is the downward institutional channel the introduction describes. We draw out what this means for moral philosophy itself in section 3.7.

3.7 Logic in Morality and its Upstream Causes

Naturalistic and evolutionary accounts of morality (Kitcher 2011; Street 2006; Joyce 2006) have challenged the view that logic is a normative requirement. They argued that moral systems are shaped by selection pressures rather than normative correspondence. What both normative and evolutionary accounts share, however, is an implicit conflation between what it means for a person to hold a moral commitment and what it means for them to hold a logically consistent one, as though rational agency and logical moral structure necessarily go together. Our account separates them.

The demand for moral consistency is one of the most successful features of the secular moral meme. It is what secular moral systems rely on in the absence of the institutional apparatus that religious moral systems have historically used to ensure propagation (Dunbar 2022, 1996; Girard 1977). This means that moral philosophers demanding logical coherence are participating in the propagation mechanism rather than evaluating it from the outside. Philosophy faculties, journals, and the tradition of moral argument are doing something functionally analogous to what the church did for religious morality: maintaining, refining, and transmitting a consistent moral code across generations. The tools are different: argument rather than ritual, publication rather than sermon, but the function is the same.

The same functional logic illuminates a familiar aspiration in legal and political philosophy: that laws should, ideally, have some moral basis. Law and morality share the same underlying function: both are coordination technologies for societies larger than personal acquaintance can sustain. Law, like morality, makes the behaviour of strangers predictable and enables cooperation without shared

personal history. The two achieve this via different means, with law using external enforcement where morality motivates people internally.

Legal systems that ground their authority in widely held moral preferences benefit from the same fitness advantages as morally structured codes: they are cheaper to communicate, more resistant to defection, and more stable over time than systems whose authority rests on enforcement alone. Where law diverges sharply from prevailing moral preferences, compensating enforcement becomes necessary, costly, contested, and fragile. The aspiration that law track morality is itself a product of selection pressure: legally codified moral preferences propagate more efficiently than either bare legal rules or free-floating moral attitudes, and the perceived moral legitimacy of law is among the most powerful amplifiers of that propagation.

3.8 Value Capture and the Metric Form of Morality

The transmission advantages developed in sections 3.5 and 3.6 are one side of a trade-off, and a complementary literature isolates the other: the cost of compression. C. Thi Nguyen's analysis of value capture (Nguyen 2024) describes how an agent who enters an environment offering a simplified, legible metric comes to adopt it in place of the richer, plural, often inarticulate values they began with, e.g. citation counts stand in for intellectual contribution, step counts for health. A quantity introduced to *represent* a value comes to *replace* it, because it is sharable, comparable, and actionable in ways the value is not. This is the individual-scale version of Scott's observation (Scott 1998) that institutions rendering a population legible must flatten the tacit, case-specific knowledge that does not survive aggregation. This is the trade-off our account locates at the heart of moral propagation, seen from its cost side: the very properties that let a moral system propagate are what make a metric legible: compressibility into a few principles, universal application without case-by-case negotiation, and overridingness that resists local trade-offs. Overridingness in particular is the moral analogue of a clear win condition, supplying what Nguyen calls value clarity (Nguyen 2020) at the price of the burden of a richer deliberation. A moral code is thus among the oldest legibility regimes.

Utilitarianism is the limiting case here. Where most moral codes acquire the metric form by degrees, it adopts it by design: a single scalar and an aggregation rule from which every verdict follows. This is exactly Scott's administrative move of replacing a heterogeneous field of values with one summable quantity. Its maximal clarity, compressibility, and decisiveness on novel cases are, on our account, fitness advantages of the first order: a theory shaped like a metric is an exceptionally good meme. But this explains its grip, not its content. The clarity that recommends it is the clarity Nguyen warns is seductive (Nguyen 2021), a property of the representation, selected for transmissibility, not a sign the representation is correct. Our framework predicts a maximally legible moral theory will propagate especially well *whether or not* its verdicts track anything.

Nguyen does not hold that all metrics should be resisted. Rather, his remedy is to match a value's form to its scale. In doing so, he keeps values thick and plural where that richness can be sustained, but grants that a shared metric's benefits outweigh the costs of flattening wherever precise collective coordination is what matters. His example is reducing carbon emissions (Nguyen 2024).

We note that in practice, large-scale normative systems remain rich and pluralist to some degree: they retain the richer form as an interpretive layer laid over the compressed one. For example, in law, a broad principle is kept open and worked out through precedent, and religious scholars dispute how deliberately vague texts should guide conduct. But this is no scalable substitute for the metric. It is invoked only where a principle buckles under a conflicting or unprecedented case, and carried out by experts within a narrow field. The faithful form is recoverable only at a cost, spent at the points where compression fails. This is why the concise principle, commandment, or statute remains the default. The seduction of clarity is therefore not a contingent error which better hygiene would correct, but the same selection pressure felt from the inside. In a broad sense, faithfulness is inversely proportional to transmissibility.

4 Objections and Responses

4.1 The Frege-Geach Problem

The most immediate technical objection to any non-cognitivist position is the Frege-Geach problem (Geach 1960, 1965). Moral statements appear not only asserted on their own but embedded in unasserted contexts, above all in the antecedents of conditionals. Ordinary reasoning treats inferences across the two as valid. Consider the inference: if killing is wrong, then getting one's brother to kill is wrong. Killing is wrong. Therefore getting one's brother to kill is wrong. For this modus ponens to be valid, "killing is wrong" must mean the same in the second premise, where it is asserted, as in the antecedent of the first, where it is only supposed and expresses no attitude. A non-cognitivist analyses the meaning of "killing is wrong" as the expression of an attitude of disapproval, yet no such attitude is expressed in that antecedent. The position therefore owes an account of how the statement keeps a constant meaning across its asserted and unasserted occurrences, and hence of why the inference is valid at all. Blackburn's quasi-realist programme (Blackburn 1984, 1993) is largely motivated by the desire to solve this problem, earning back the right to talk as if moral statements behave logically, starting from an expressivist base.

We do not attempt to solve the Frege-Geach problem, and we do not think we need to. On our account the occurrence of "killing is wrong" in the antecedent expresses no attitude, so the inference does not carry anything from premises to conclusion the way a truth-preserving inference carries truth. Its apparent validity is not genuine semantic entailment, and we accept this. This is of a piece with a more general feature of moral commitment on our view, namely that consistency is not binding on it: someone can hold that killing is always wrong and also that they did it this one time and that this was morally OK, with no logical contradiction following, since a moral preference is not beholden to a specific logic. Rather than treating any of this as a deficiency that requires repair, we treat it as reflecting the actual nature of moral statements. This also means that the Frege-Geach problem, while real as a problem for quasi-realism, is not a problem for our position. It is a problem for anyone who wants to combine non-cognitivism with the semantic compositionality that logical inference requires.

When a person embeds a moral attitude in a conditional argument of the form “if killing is wrong and this is killing, then this is wrong,” they are not performing semantic compositionality. They are executing a rhetorical manoeuvre: borrowing the psychological authority of deductive form to make a moral attitude harder to resist. Our account explains why this pattern became so prevalent in moral discourse. Arguments structured as deductions are easier to transmit, harder to dismiss without explicit counter-argument, and more resistant to defection than bare attitude expressions. The Frege-Geach problem identifies what looks like a logical defect in non-cognitivism. On our account, it identifies a structural feature that moral discourse has learned to exploit.

This reading is not uncharitable to moral discourse, since the same deductive form has a cooperative function alongside the exploit one: rendering a preference as “if killing is wrong and this is killing, then this is wrong” makes the underlying commitments inspectable, so a competent hearer can locate them, weigh them against their own, and challenge them at specific points rather than merely be swayed. The embedding thus does two things at once, and which dominates depends on whether it is used to foreclose disagreement or to enable it.

4.2 The Self-Undermining Worry

A reflexive objection presses on any account of this kind. If moral reasoning is, as we claim, a faculty for persuasion and propagation rather than truth-tracking, then the reasoning in this paper is open to the same diagnosis, and a reader has no more reason to accept it than any other persuasive performance. The account would appear to undermine itself.

The first reply is that this misidentifies our project. The study of moral codes themselves and the action guidance they provide through logical reasoning and application thereof to specific situations presupposes the logical normativity of moral codes. The claim we defend is metaethical and descriptive rather than prescriptive: it describes which moral attitudes people in fact hold, why the ones that persist tend to be logically structured, for the reasons already given, and how those attitudes bear on social life, how and when they are applied, and how they came to be. It does not urge anyone to adopt a first-order moral code. What the account says about the propagation of moral

preferences therefore does not turn on itself, because the argument is not itself a bid to propagate a moral preference.

A second reply reinforces the first. Even granting Mercier and Sperber's claim that reasoning evolved as a faculty for persuasion, the modern development of formal and informal logic supplies public standards against which an argument can be assessed with some independence from the interests of whoever advances it. Mercier and Sperber themselves hold that argumentation is truth-conducive in just this dialogical setting, where the production of arguments is biased but their evaluation by others is not. The present account is offered in that spirit, as a claim to be assessed against shared premises about cultural transmission and the scaling of cooperation, not as a bid to propagate a moral code.

4.3 Debunking and Relativism

A natural objection to any genealogical account of morality is that it debunks: if we can explain why people hold moral beliefs by appeal to selection pressures rather than truth-tracking, we have undermined their standing. Our position escapes this, and for the same reason in every version of the worry. We are non-cognitivists: moral statements express preferences, not claims about moral facts, and so are not truth-apt to begin with. A genealogy can debunk a belief, by showing it was not produced by a truth-tracking process, but it cannot debunk a preference by that move, because preferences were never in the business of tracking truth. Showing that a preference was shaped by selection pressure leaves its standing as a preference entirely intact.

This is what separates us from the error theorists. Mackie (Mackie 1977) and Joyce (Joyce 2006) hold that moral statements purport to describe objective moral facts that do not exist and are therefore systematically false, but that charge lands only on a view that takes them to be truth-apt, and ours does not. The non-cognitivist alternative we occupy is precisely the one Joyce's analysis makes room for but declines. Street's (2006) Darwinian dilemma is likewise aimed at realists, holding that evolved evaluative attitudes either implausibly track independent moral facts or are severed from moral truth altogether.

The account also differs from Nietzsche's (1887). He took genealogy to loosen the grip of values by exposing them as the sediment of power struggle and resentment. On our account, on the other hand, it has no such effect, since a preference shaped by propagation is still a genuine preference, worth holding and acting on, and the history of how it got there does not change what it is. What genealogy leaves open is not liberation but deliberate revision: we remain free to choose which preferences to keep, which to adjust, and whether to seek consistency among them.

A related worry is that the account invites relativism (thereby undermining constructivism): if moral claims are preferences rather than truths and the systems that survive are simply those that propagate, then goodness seems to reduce to memetic fitness, with no standpoint from which one culture's morality is better than another's. But fitness is not goodness. Selection favours transmissibility and in-group coordination, not the welfare of everyone a system affects, and a coherent, universalisable code can spread while harming outsiders, so nothing here licenses the inference from what propagates to what is good. What remains available is comparative evaluation: the criteria one would use, welfare, coordination, and stability, are constructed within moral cultures rather than read off an external standard, but from within any such framework a system's effects are empirically assessable, and one that consistently produces collapse tends not to persist. The evaluation is thus relative to a framework but not arbitrary, and since the attitudes any new moral meme must fit were themselves shaped by earlier ones, there is no neutral baseline to demand in its place. The relativist's observation that a morality fits the culture that holds it is then not a rival thesis but one instance of the fitness mechanism at work: how well a system fits the attitudes already in place is part of its memetic fitness (for which its logical structure matters, as we argue here). So, because those attitudes differ from one culture to the next, so does what can propagate.

5 Conclusion

The argument of this paper is metaethical and descriptive in character. We have not argued that any particular moral code is correct, or that morality ought to be logical, but that the logical structure typically demanded of moral systems, its consistency, universalisability, and overridingness, is best

understood as a memetic fitness advantage, a property moral systems tend to acquire because those that have it outcompete those that do not. That structure earns its place by doing three things: it makes moral reasoning and discourse possible, it lets an agent and a community pre-commit so that behaviour stays predictable, and it makes moral systems transmissible across persons and generations while coordinating strangers at scale.

The account reaches this conclusion by an interdisciplinary synthesis. Mercier and Sperber's account of reasoning as a social persuasion faculty supplies the mechanism; Dunbar's findings on the scaling of human social groups identify the environment in which logical structure becomes fitness-decisive; and the empirical moral psychology shows that reasoning typically follows judgement rather than producing it, so that Hare's logical constraints reappear not as norms but as the structural properties selection acts on. Logical structure is in this sense the machinery of moral transmission, or more precisely a side-effect of the machinery of persuasion. Several things follow from this vantage. Read alongside Scott on legibility and Nguyen on value capture, the same structure is what turns a moral code into a legible metric, with utilitarianism as the limiting case: a morality in metric form whose clarity is a fitness advantage rather than a mark of truth; and the Frege-Geach problem reappears not as a semantic defect non-cognitivism must repair but as a structural pattern moral discourse exploits for transmission and uses to make preferences publicly inspectable.

The account also explains a peculiarity of moral philosophy itself. Among systematic inquiries it is unusual in being largely prescriptive rather than descriptive, advocating codes and telling people how they ought to act. On our account this is what one would expect: the demand for consistency is a move in the propagation of a moral system, not a discovery about a normative order (section 3.7). The prescriptive posture of ethics is therefore itself a data point our descriptive metaethics explains, not a rival to it.

Several limitations deserve acknowledgement. The empirical work we have drawn on, namely Dunbar's anthropological data, the moral psychology literature, and Sperber's epidemiology of representations, is used here as theoretical support for a philosophical argument rather than as direct

evidence for claims about secular moral propagation specifically. A fuller account would ideally be grounded in systematic sociological study: data on how secular moral frameworks propagate across populations, how their degree of logical structure correlates with their durability and reach, and how institutions such as schools, universities, legal institutions, workplaces, and media mediate propagation in the absence of a religious apparatus. Such work is methodologically demanding, since secular moral preferences are harder to identify and track empirically than religious affiliation or doctrinal commitment, but it represents the natural empirical development of the theoretical argument made here.

A further qualification concerns the scope of the account. It assumes a degree of homogeneity in the community of moral discourse that is not obtainable in practice. Different linguistic and cultural communities have different colloquial understandings of how logic applies to moral reasoning, and different starting axioms within their moral codes. Where these underlying frameworks diverge, whether across languages, cultural traditions, or forms of moral community, cross-community moral communication encounters a distinctive kind of friction. The moral meme must fit existing cognitive and discursive dispositions in order to propagate. Where those dispositions differ substantially, even a logically coherent moral system may fail to cross communal boundaries. The fitness advantage of logical structure is therefore context-dependent.

Three directions remain for future work. The first is that some moral memes appear to propagate not only through the fitness mechanisms described above but also through cost-imposition that specifically tracks resistance to the moral view. Doctrines of damnation for the unfaithful, shame and exclusion attached to dissent, and guilt mechanisms tied to specific moral failures all display a structural signature in which the cost to the target intensifies when they attempt to maintain a contrary view and eases when they yield. This is the signature of manipulation rather than mere transmission. The relationship between propagation pressure as analysed here and manipulation as a specific propagation mechanism is a natural extension of the present account.

The second runs in the opposite direction to the core fitness story, and is worth study for the tension it creates: logical strain in a moral framework does not always weaken it. Where holding

a demanding, costly, or even internally contradictory belief is itself the price of membership, the strain can bind adherents more tightly rather than driving them away, since visibly paying that price is a hard-to-fake signal of commitment that outsiders and defectors will not match. The escalating and increasingly counterintuitive doctrines that adherents of some high-control groups are expected to hold as they rise through the ranks are an extreme case, in which the very implausibility of the belief is what makes professing it a credible display of loyalty, and the effort already sunk into the framework raises the cost of leaving. This suggests that the pressure toward logical structure is not the only one operating. A rival pressure, toward costly and identity-constituting commitment, can favour precisely the friction the first would smooth away, and mapping the conditions under which each dominates is a natural extension of the account.

The third concerns the extension of the moral domain rather than its structure. In this work, we have taken the class of moral preferences as given. However, which conduct a community moralises, as opposed to leaving it to taste, prudence, or convention, is itself variable and historically mobile: intelligence, dress, food, beauty, and sexual conduct fall inside the moral domain in some cultures and outside it in others, and the boundary moves within a single culture over time, as smoking and meat-eating acquired the moral status that homosexuality and divorce lost (Rozin 1999; Rozin et al. 1997). The present account has resources to address this. Coordination-relevant conduct is moralised because attaching universalisability and overridingness to it is what makes it coordination-effective, while purity, dress, and dietary norms, coordination-irrelevant on their face, plausibly moralise as group-identification markers whose policing sharpens the boundary of the cooperating unit (Norenzayan 2013; Henrich 2020). On this view the boundary of the moral domain would fall under the same selection as its logical structure, its shifts through moralisation and demoralisation tracking changes in the coordination or signalling value of the conduct in question, and the properties that make a norm propagate would be the very marks by which a moralised value is distinguished from a bare preference.

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